

GRASP: Reinforcing Language Model Anonymizers with Group Relative Policy Optimization

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Abstract

Large language models can infer sensitive personal attributes, such as age, location, and occupation, from ordinary text, turning everyday writing into a privacy risk. Adversarial anonymization defends against this by rewriting a text with a capable language model that also plays the attacker, but it needs a powerful model at inference time and thus sends private text to a third party, the very exposure anonymization should prevent. Recent work distills this behavior into a small on-device model using supervised fine-tuning and direct preference optimization (DPO), but DPO only imitates the teacher’s offline choices and never directly optimizes the privacy–utility objective we care about. We introduce **GRASP** (Group-Relative Anonymization via Self-refinement Policy-optimization), which reinforces the local anonymizer online with Group Relative Policy Optimization. A single small model acts as anonymizer, adversary, and utility judge, trained against a self-generated reward that hides attributes while preserving meaning, with a design that guards against reward hacking. Trained on Llama-3.1-8B, GRASP improves the privacy–utility trade-off over the DPO-distilled baseline, consistently across three independent LLM judges. Against adversarial anonymization driven by frontier models such as Gemini 2.5 Flash and Claude, it achieves a comparable or better overall trade-off while removing substantially more private information, and it runs entirely on-device at roughly 1% of the GPT-4o teacher’s cost.

1 Introduction

Large language models can infer sensitive personal attributes, such as age, location, occupation, and income, from ordinary text with surprising accuracy (Staab et al., 2024, 2025), turning routine online writing into a privacy liability and compounding known exposures such as memorization and ex-traction of training data (Carlini et al., 2021, 2022).

As people increasingly write with and share text through LLMs (Achiam et al., 2023; Grattafiori et al., 2024; Team et al., 2025), there is a growing need to rewrite text so that its meaning is preserved while private attributes can no longer be inferred (Dou et al., 2024).

Classical anonymization removes explicit identifiers through named-entity recognition or rule-based redaction (Microsoft, 2025; Pilán et al., 2022), and paraphrasing models alter surface form (Krishna et al., 2023), but neither addresses the contextual cues that modern inference attacks exploit (Gadotti et al., 2024). The strongest defense to date is adversarial anonymization (Staab et al., 2025; Yang et al., 2025; Frikha et al., 2025), which iteratively rewrites a text using a capable LLM that also plays the attacker. This is effective but requires a powerful model at inference time, so users must either pay for a frontier API or send private text to a third party (Hou et al., 2025), which is precisely the exposure that anonymization is meant to prevent.

To keep anonymization local, SEAL (Kim et al., 2026) distills (Hinton et al., 2015; Levine et al., 2025) adversarial anonymization into a small language model that can anonymize, infer, and evaluate its own rewrites, using supervised fine-tuning followed by direct preference optimization (DPO) (Rafailov et al., 2023). DPO, however, only imitates preferences mined offline from the teacher’s trajectories: it learns to reproduce which rewrites the teacher preferred, but never directly optimizes the privacy–utility objective we care about, and it cannot improve beyond the demonstrations it distilled from.

Our key observation is that anonymization comes with a well-defined and measurable objective, namely whether an adversary can still infer the target attributes and whether the text’s utility is preserved, and that a single distilled model can act as anonymizer, adversary, and utility judge. This makes it natural to optimize the objective *directly*

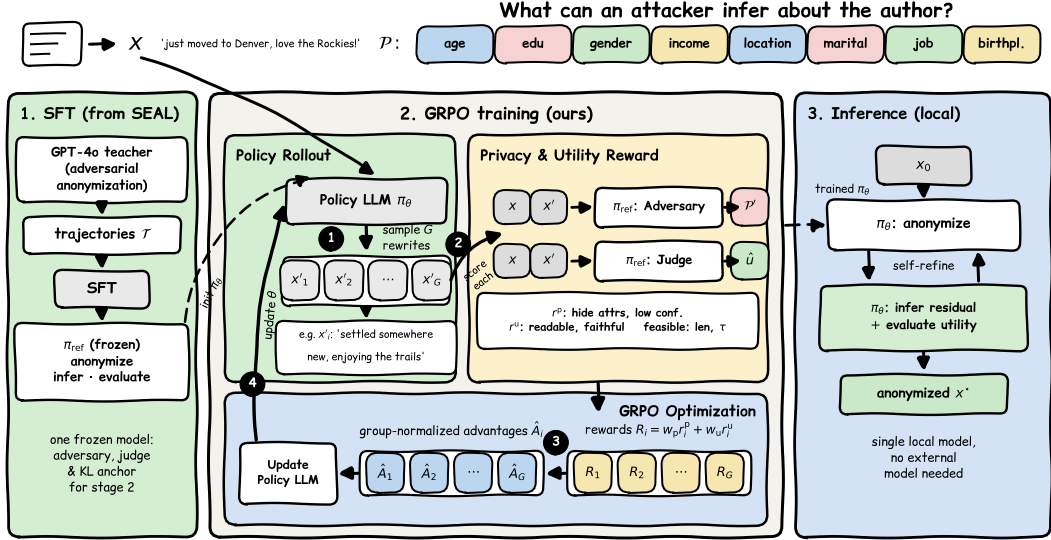


Figure 1: **Overview of GRASP.** *Stage 1:* we reuse SEAL’s GPT-4o adversarial-anonymization trajectories to supervise-fine-tune a small language model into a frozen reference π_{ref} that can anonymize, infer attributes, and evaluate utility. *Stage 2:* we optimize the anonymization policy π_θ with GRPO. For each text, π_θ samples a group of candidate rewrites; the frozen π_{ref} scores each as adversary and judge into a privacy–utility reward, and group-relative advantages update the policy with a KL anchor to π_{ref} . *Stage 3:* at inference the trained policy runs entirely on-device, self-refining by alternately anonymizing and critiquing its own output, with no external model.

with online reinforcement learning rather than imitate a teacher offline. Whereas reinforcement learning from human feedback trains language models against costly human preference labels (Christiano et al., 2017; Ziegler et al., 2019; Stiennon et al., 2020; Ouyang et al., 2022), our reward is generated by the model itself, as it iteratively refines its own rewrites (Madaan et al., 2023) and scores them as an attacker and an LLM judge (Zheng et al., 2023).

We introduce **GRASP** (Group-Relative Anonymization via Self-refinement Policy-optimization), which fine-tunes the anonymization policy with Group Relative Policy Optimization (GRPO) (Shao et al., 2024; Guo et al., 2025) against a reward that combines the residual attribute inference of a frozen reference model with its utility assessment. Unlike PPO (Schulman et al., 2017), GRPO estimates advantages from a group of sampled rewrites and needs no separately learned value network, which suits variable-length text generation and maps naturally onto comparing several candidate anonymizations of the same input. Because the policy is scored on its own samples, it explores rewrites the teacher never produced and pushes the privacy–utility frontier past the distilled ceiling. The same frozen model supplies the reward and anchors the update, so no external model is needed at training or inference.

Figure 1 illustrates the full pipeline.

Trained on Llama-3.1-8B (Grattafiori et al., 2024), GRASP attains a stronger privacy–utility trade-off than the state-of-the-art DPO-distilled baseline, consistently across three independent LLM judges. Against adversarial anonymization driven by frontier models such as Gemini 2.5 Flash (Google DeepMind, 2025) and Claude (Anthropic, 2025), it achieves a comparable or better overall trade-off while removing substantially more private information, all while running entirely on-device at roughly 1% of the cost of the GPT-4o (Achiam et al., 2023) teacher from which it is distilled, and well below that of the frontier anonymizers. In summary, our contributions are: (1) we cast local text anonymization as direct reinforcement learning against a self-generated adversarial reward, rather than offline imitation of a teacher; (2) we implement GRASP, a GRPO recipe in which one small model serves as anonymizer, attacker, and utility judge; and (3) through extensive experiments and ablations we show that GRASP improves on the privacy–utility trade-off over distillation, and over frontier-model baselines in privacy, while remaining cheap and fully local.

2 Preliminaries

We study anonymization against *inference-time* privacy threats, in which a capable language model acts as an adversary that infers private attributes from seemingly innocuous text (Staab et al., 2024). This section formalizes the threat model and reviews the adversarial anonymization procedure and the trajectory data on which our method builds.

Threat model and problem statement. Let x be a text written by a user that may leak private attributes $\mathcal{P}$ (e.g., age, gender, location, occupation) about its author. An adversary is a language model $\mathcal{M}_{\text{priv}}$ that, given x and a target attribute set $\mathcal{P}$, predicts each attribute’s value with a confidence score. Unlike classical identifiers removed by named-entity recognition or pattern matching, these attributes are often encoded in context-dependent semantic cues that survive surface-level scrubbing (Staab et al., 2024). An anonymizer rewrites x into x' so that the attributes in $\mathcal{P}$ can no longer be reliably inferred from x' , while preserving the original semantic content and readability. The goal is thus a favorable *privacy–utility trade-off*: strong protection against attribute inference at minimal cost to the meaning and fluency of the text.

Adversarial anonymization. A powerful way to obtain such rewrites is feedback-guided adversarial anonymization (Staab et al., 2025), in which an anonymizer $\mathcal{M}_{\text{anon}}$ and an adversary $\mathcal{M}_{\text{priv}}$ alternate: the adversary infers the attributes that remain exposed, and the anonymizer edits the text to remove them. Given an initial text x_0 and target attributes $\mathcal{P}$, and additionally introducing a utility evaluator $\mathcal{M}_{\text{util}}$, the procedure repeats the following three steps for $t = 0, 1, \dots, T - 1$: **(1) inference** $\mathcal{P}_t \sim \mathcal{M}_{\text{priv}}(x_t, \mathcal{P})$, **(2) refinement** $x_{t+1} \sim \mathcal{M}_{\text{anon}}(x_t, \mathcal{P}_t)$, and **(3) utility** $\mathcal{U}_{t+1} \sim \mathcal{M}_{\text{util}}(x_0, x_{t+1})$, where each inferred attribute $p \in \mathcal{P}_t$ carries a rationale and a confidence score $\text{conf}(p)$, and $\mathcal{U}_{t+1}$ collects utility measures (e.g., readability and semantic preservation) of x_{t+1} relative to x_0 . The procedure terminates after a fixed number of steps or once no further attributes can be inferred, yielding a trajectory $\tau = (s_0, s_1, \dots, s_T)$, where each state $s_i = (x_i, \mathcal{P}_i, \mathcal{U}_i)$ pairs a text with its inferred attributes and utility measurements. Running many LLMs as $\mathcal{M}_{\text{anon}}$, $\mathcal{M}_{\text{priv}}$, and $\mathcal{M}_{\text{util}}$ over a corpus produces a trajectory set $\mathcal{T}$ capturing how a text can be progressively anonymized (Staab et al., 2025).

Scoring anonymizations. To compare states within a trajectory we use scalar privacy and utility scores. Following Staab et al. (2025), an anonymization is more private when fewer attributes are inferred and those that are inferred carry lower confidence, and it is more useful when its utility measures are higher on average:

$$p(s_i) = \left(-|\mathcal{P}_i|, -\frac{1}{|\mathcal{P}_i|} \sum_{m \in \mathcal{P}_i} \text{conf}(m) \right), \quad (1)$$

$$u(s_i) = \frac{1}{|\mathcal{U}_i|} \sum_{m \in \mathcal{U}_i} m. \quad (2)$$

Privacy scores are ordered lexicographically (first by the number of inferred attributes, then by mean confidence), so that $p(s_j) > p(s_i)$ means s_j is strictly more private than s_i ; utility scores are compared by their average. The orderings identify, within each trajectory, which later rewrites dominate earlier ones in privacy and utility, and form the supervision signal for distilling anonymization behavior into a small language model.

Distillation setup. Our aim is to distill this behavior into a single small language model (SLM) that performs anonymization and its own evaluation, so that anonymization can run locally without querying external, potentially untrusted models (Staab et al., 2025). Concretely, given the trajectory set $\mathcal{T}$ collected above, we train a target policy π_θ that (a) rewrites a text to improve its privacy–utility trade-off, (b) infers private attributes from a text as an adversary, and (c) evaluates the utility of a rewrite. Section 3 describes how we adapt the model to these tasks and then optimize its anonymization policy with reinforcement learning.

Task adaptation via supervised fine-tuning. We adapt the base model to the three abilities for self-refinement: (i) *anonymization*, rewriting a text to reduce attribute inferability; (ii) *adversarial inference*, predicting private attributes with confidences; and (iii) *utility evaluation*, judging semantic preservation and readability of a rewrite. Using the privacy and utility orderings defined above, we mine, from each trajectory $\tau \in \mathcal{T}$, the pairs in which a later rewrite dominates an earlier one, together with per-state inference and utility labels:

$$\begin{aligned} \mathcal{D}_{\text{anon}} &= \{(x_i, x_j) \mid 0 \leq i < j \leq T, \\ &\quad p(s_j) > p(s_i), u(s_j) \geq u(s_i)\}, \\ \mathcal{D}_{\text{priv}} &= \{(x_i, \mathcal{P}_i) \mid s_i \in \tau\}, \\ \mathcal{D}_{\text{util}} &= \{(x_i, \mathcal{U}_i) \mid s_i \in \tau\}. \end{aligned}$$

The model π_θ is trained to minimize a weighted sum of next-token losses over the three datasets,

$$\mathcal{L}_{\text{SFT}} = \lambda_{\text{anon}} \mathcal{L}_{\text{anon}}(\mathcal{D}_{\text{anon}}) + \lambda_{\text{priv}} \mathcal{L}_{\text{priv}}(\mathcal{D}_{\text{priv}}) + \lambda_{\text{util}} \mathcal{L}_{\text{util}}(\mathcal{D}_{\text{util}}),$$

so that a single model can both generate and evaluate anonymizations. We denote the resulting model π_{ref} ; it serves as the initialization for the reinforcement-learning stage and as the frozen reference used in its reward and policy regularizer.

3 Proposed Algorithm

Our method trains a single small language model to be both a strong anonymizer and its own evaluator. We build on the supervised fine-tuned model π_{ref} of Section 2, which adapts the base model to anonymization, adversarial inference, and utility evaluation following the SEAL distillation recipe. Our central contribution is to replace the preference-learning (DPO) step of the prior distillation approach (Kim et al., 2026) with reinforcement learning that directly optimizes the privacy–utility trade-off. We fine-tune the anonymization policy with Group Relative Policy Optimization (GRPO) (Shao et al., 2024) against an explicit reward, using the frozen reference model π_{ref} itself as the adversary and utility evaluator.

3.1 Policy Optimization for Anonymization via GRPO

Because SFT only imitates trajectory data, we fine-tune the anonymization policy with reinforcement learning, treating each rewrite as an action and scoring it with a reward that encodes both privacy and utility. Algorithm 1 summarizes the procedure.

Reward design. Given an input text x with target attribute set $\mathcal{P}$, the policy proposes an anonymization $x' \sim \pi_\theta(\cdot | x)$. We score x' with two components. For *privacy*, the reference model, acting as the adversary $\mathcal{M}_{\text{priv}} = \pi_{\text{ref}}$, attempts to infer the target attributes from x' ; let $\mathcal{P}' \subseteq \mathcal{P}$ be the attributes for which it produces a confident guess, each with confidence $\text{conf}(m)$. We count *any* confident inference, irrespective of whether the guess is correct (we return to this choice below). The privacy reward then rewards hiding attributes and lowering the adversary’s confidence,

$$r_{\text{p}}(x') = \alpha \left(1 - \frac{|\mathcal{P}'|}{|\mathcal{P}|} \right) + \beta \left(1 - \frac{1}{|\mathcal{P}'|} \sum_{m \in \mathcal{P}'} \text{conf}(m) \right), \quad (3)$$

with $\alpha + \beta = 1$ and the convention that the confidence term equals 1 when $\mathcal{P}' = \emptyset$ (so a fully anonymized rewrite attains $r_{\text{p}} = 1$). For *utility*, the same reference model acts as the evaluator $\mathcal{M}_{\text{util}} = \pi_{\text{ref}}$, comparing x' to the original x and returning normalized scores for readability, semantic preservation, and freedom from hallucinated content, whose average we denote $r_{\text{u}}(x') \in [0, 1]$. The two components are combined subject to two hard constraints that rule out degenerate rewrites:

$$R(x') = \begin{cases} w_{\text{p}} r_{\text{p}}(x') + w_{\text{u}} r_{\text{u}}(x'), & x' \text{ feasible,} \\ -1, & \text{otherwise,} \end{cases} \quad (4)$$

where a rewrite x' is *feasible* when it clears a utility floor, $r_{\text{u}}(x') \geq \tau$, and stays within a length band, $|x'| \in [\frac{1}{2}|x|, 2|x|]$ in words. The length constraint discourages trivial solutions that delete or pad the text, and the utility floor τ prevents the policy from purchasing privacy at the cost of destroying meaning; w_{p} and w_{u} set the privacy–utility balance.

Design choices against reward hacking. The challenging part is not switching from preference learning to GRPO, but making the on-policy optimization *stable*. Because GRPO trains the policy on its own samples, the policy will exploit any weaknesses in the reward instead of learning the intended behavior. Each part of the reward addresses a specific failure we encountered during training.

- **Confidence makes the signal learnable.** Counting only how many attributes stay exposed is a coarse, almost all-or-nothing score, so rewrites in a group often tie, and GRPO finds no difference to learn from. The continuous confidence term rewards partial progress, separating a rewrite that merely lowers the adversary’s certainty from one that removes a cue outright. Without it, the model stops hiding information and privacy falls back to nearly the original level (ablations are presented in Table 3).
- **Penalize any confident guess, not only correct ones.** A truth-aligned reward that only penalizes the adversary for being *right* can be satisfied without removing information: the policy keeps the revealing cue but rewords it just enough to make the reference attacker guess wrong, leaving the private detail in place, only disguised. A stronger evaluation attacker sees through the disguise and recovers it, so real privacy does not improve (truth-aligned reward, Table 3). Pe-

nalizing *any* confident guess instead forces the policy to delete the cue.

- **The length band blocks empty rewrites.** With no length limit, the policy discovers that a very short, near-empty rewrite leaks nothing and scores high on privacy while destroying the content. Requiring the output to stay within $[\frac{1}{2}|x|, 2|x|]$ words removes this shortcut.
- **The utility floor keeps text usable.** Without a floor, the policy keeps trading meaning for privacy over successive self-refinement rounds and utility drops sharply. The floor $r_u \geq \tau$ keeps every rewrite readable and faithful, giving a better-balanced result ($\tau=0$, Table 3).

Reference model as adversary. A key design choice is that both $\mathcal{M}_{\text{priv}}$ and $\mathcal{M}_{\text{util}}$ in the reward are instantiated by the frozen reference model π_{ref} from Section 2, rather than a separately trained attacker or critic. Because SFT already equips the model with adversarial-inference and utility-evaluation abilities, π_{ref} supplies the reward, and the same frozen model serves as the KL anchor of the policy update below. This keeps training self-contained and avoids a second network.

Our GRPO objective. We optimize the policy with GRPO (Shao et al., 2024), which estimates advantages from *groups* of sampled outputs and therefore requires no learned value network, unlike PPO (Schulman et al., 2017). For each input x , we draw a group of G anonymizations $\{x'_1, \dots, x'_G\} \sim \pi_{\theta_{\text{old}}}(\cdot | x)$, score each with $R_i = R(x'_i)$ from Eq. (4), and normalize the rewards within the group to form advantages $\hat{A}_i = \frac{R_i - \text{mean}(\{R_j\}_{j=1}^G)}{\text{std}(\{R_j\}_{j=1}^G)}$, which are shared across all tokens of x'_i . The policy is updated by maximizing the clipped surrogate objective with a KL penalty to the reference model,

$$\mathcal{J}(\theta) = \mathbb{E} \left[\frac{1}{G} \sum_i \frac{1}{|x'_i|} \sum_t \ell_{i,t}(\theta) \right] - \beta_{\text{KL}} \mathbb{D}_{\text{KL}}(\pi_{\theta} \parallel \pi_{\text{ref}}), \quad (5)$$

where the outer average runs over the G group samples and the inner over the tokens of x'_i , and the per-token clipped surrogate is

$$\ell_{i,t}(\theta) = \min(\rho_{i,t} \hat{A}_i, \text{clip}(\rho_{i,t}, 1-\epsilon, 1+\epsilon) \hat{A}_i).$$

Here ϵ is the clipping range, β_{KL} the strength of the reference regularizer, and the token-level importance ratio is $\rho_{i,t} = \frac{\pi_{\theta}(x'_{i,t} | x, x'_{i,<t})}{\pi_{\theta_{\text{old}}}(x'_{i,t} | x, x'_{i,<t})}$. Intuitively,

anonymizations that beat their group-mates on the privacy–utility reward are up-weighted, while the KL term keeps the policy close to the SFT reference and preserves fluency.

3.2 Iterative Self-Refinement at Inference

At test time the trained policy anonymizes by self-refinement, using no external model. Starting from x_0 , at each step t the model infers the residual private attributes $\mathcal{P}_t^{\pi}$ and evaluates the utility $\mathcal{U}_t^{\pi}$ of the current text, then produces a refined rewrite conditioned on its own feedback,

$$x_{t+1} \sim \pi_{\theta}(\cdot | x_t, \mathcal{P}_t^{\pi}, \mathcal{U}_t^{\pi}).$$

The loop continues until a target privacy–utility trade-off is reached or a fixed iteration budget is exhausted. Because a single local model performs both generation and critique, anonymization requires no proprietary or external adversary, and users can steer the number of refinement rounds to trade privacy against utility as they prefer.

4 Empirical Results

4.1 Setup

Datasets. We evaluate on SynthPAI (Yukhy-menko et al., 2024), a corpus of synthetic personal profiles paired with text comments labeled for eight personal attributes (age, education level, gender, income level, location, marital status, occupation, and place of birth). We use the anonymization trajectories of the SEAL distillation baseline, produced by adversarial anonymization (Staab et al., 2025) with GPT-4o (Achiam et al., 2023). We evaluate on two held-out splits never seen in training: a *main* set of 723 comments and a *hard* set of 500 texts that embed personal information contextually rather than as explicit identifiers, testing generalization to harder, unseen cases. More details on the dataset can be found in Appendix A.

Baselines. We compare GRASP against the original (unmodified) text and several anonymization methods. As a rule-based reference we include Azure’s PII detection tool (Microsoft, 2025), which redacts sensitive spans via named-entity recognition. As a pure semantic-rewriting reference we include Dipper (Krishna et al., 2023), an 11B paraphraser not designed for anonymization. As the strongest prior LLM-based method we include adversarial anonymization (Staab et al., 2025), which iteratively rewrites

text through interaction between an anonymizer and an inference model, run with three frontier anonymizers (Gemini 2.5 Flash (Google DeepMind, 2025), Claude Haiku 4.5 (Anthropic, 2025), and Qwen3 (Qwen Team, 2025)). Finally, we compare against SEAL (Kim et al., 2026), the SFT+DPO distillation baseline, which is the most direct point of comparison for isolating the effect of reinforcement learning.

Evaluating privacy and utility. To measure privacy, we use GLM-5 (Zhipu AI, 2026) as the inference model. Since capable LLMs are strong attribute-inference attackers (Staab et al., 2024), adopting a powerful model here makes the privacy evaluation stringent. The model performs zero-shot chain-of-thought inference over the eight attributes, and we report attribute-inference accuracy, the average fraction of correctly inferred attributes, so that lower is better. For utility, we use a GLM-5 judge to score each anonymization on (1) readability, (2) semantic preservation relative to the input, and (3) absence of hallucinated content, and we take their average as the utility score. Following SEAL, we summarize the trade-off with an *overall* score, the relative privacy improvement minus the relative utility loss, normalized by the original privacy. To confirm that our conclusions are not an artifact of a single evaluator, we additionally repeat the assessment with two independent judges, Qwen3 and GPT-OSS (Agarwal et al., 2025) (Section 4.3).

Implementation details. We train GRASP on Llama-3.1-8B-Instruct (Grattafiori et al., 2024). The SFT stage follows the distillation baseline exactly, using LoRA with rank 16, $\alpha=16$, and dropout 0.05, AdamW at learning rate $2e-4$, and one epoch. We then initialize the policy from the SFT model, which also serves as the frozen reference π_{ref} , and optimize it with GRPO for one epoch on 5,000 training prompts. Each prompt draws $G=8$ sampled anonymizations, and training uses learning rate $5e-6$, KL weight $\beta_{\text{KL}}=0.03$, effective batch size 32, sampling temperature 0.9, and a maximum completion length of 512 tokens. The reward uses privacy coefficients $\alpha=0.7$, $\beta=0.3$, trade-off weights $w_p=0.9$, $w_u=0.1$, and utility floor $\tau=0.55$. All models are trained with LoRA (Hu et al., 2022) on NVIDIA GPUs. At inference the trained policy anonymizes by self-refinement for up to five iterations. Table 5 summarizes all hyperparameters for both stages.

4.2 Overall Performance

Main dataset. Table 1 reports results on the main dataset. Among the baselines, SEAL is the strongest (overall 0.368), ahead of adversarial anonymization with frontier models (Claude 0.292, Gemini 0.284, Qwen3 0.207), while the rule-based Azure and the paraphrasing Dipper barely reduce attribute inference and push the overall score below zero. GRASP improves on all of them. At its best-Overall round it reaches **0.396**, reducing attribute-inference accuracy to 0.260—a 29% relative reduction from the best frontier anonymizer (Claude, 0.365)—at a 7% relative utility cost (0.815 vs. 0.880). With continued self-refinement, GRASP reaches inference accuracy 0.195, below SEAL’s most private round (0.244) and the lowest of any method, while keeping readability at 0.982. Its advantage over SEAL is concentrated on contextually encoded attributes: at the most private round it lowers residual inference on age (0.300 vs. 0.500), marital status (0.182 vs. 0.425), gender (0.250 vs. 0.417), and education (0.203 vs. 0.325), while both methods drive location and occupation close to zero. As Fig. 2 shows, GRASP traces the outer privacy–utility frontier on this split, whereas Azure and Dipper yield little privacy gain and Dipper incurs the largest utility loss.

Hard dataset. On the harder set of texts with contextually embedded personal information, all methods remove fewer attributes (Table 2). The frontier adversarial-anonymization models are more competitive on the aggregate score, with Gemini (0.301) and Claude (0.299) narrowly ahead of GRASP’s best overall of 0.294. This parity, however, reflects a different operating point rather than comparable anonymization: the frontier models preserve utility by stopping far short on privacy, leaving inference accuracy between 0.454 and 0.517, whereas GRASP continues to reduce it to 0.332 with further refinement, the lowest of any method on this split. GRASP also improves on the SEAL baseline (0.294 vs. 0.266 overall); as on the main split, its advantage over SEAL is largest on education (0.417 vs. 0.512 at the most private round), gender (0.500 vs. 0.619), marital status (0.527 vs. 0.613), and occupation (0.117 vs. 0.147). Utility remains comparable to the main split, with readability above 0.93 at every round. As Fig. 2 shows, GRASP extends the privacy–utility frontier on this split as well, reaching inference accuracies no baseline attains at any utility level.

Table 1: **Anonymization on the main dataset**, judged by GLM-5. For SEAL and GRASP we report the best-Overall (_O) and best-privacy (_P) rounds; full trajectories are in Table 8. **Bold/underline**: best/second-best per row.

Metric	Original	Azure	Dipper	Adv. Anon.			SEAL		GRASP (Ours)	
				Gemini	Claude	Qwen3	best-O	best-P	best-O	best-P
Overall ↑	-	-0.033	-0.018	0.284	0.292	0.207	0.368	0.304	0.396	<u>0.368</u>
Privacy ↓	0.620	0.610	0.538	0.367	0.365	0.454	0.262	<u>0.244</u>	0.260	0.195
Age	0.762	0.802	0.614	0.720	0.614	0.750	0.551	<u>0.500</u>	0.600	0.300
Edu	0.645	0.667	0.718	0.521	0.510	0.576	0.416	<u>0.325</u>	<u>0.266</u>	0.203
Gnd	0.883	0.820	0.721	0.508	0.508	0.590	0.623	0.450	<u>0.417</u>	0.250
Inc	0.561	0.598	0.531	0.510	<u>0.490</u>	0.561	0.457	<u>0.521</u>	0.607	0.536
Loc	0.470	0.362	0.300	0.067	0.090	0.097	0.015	<u>0.023</u>	0.050	0.050
Mar	0.635	0.676	0.541	0.554	0.608	0.693	<u>0.324</u>	<u>0.425</u>	0.409	0.182
Occ	0.578	0.542	0.448	0.163	0.167	0.289	0.289	0.048	<u>0.069</u>	0.083
PoB	0.321	0.286	0.167	0.143	0.179	0.143	0.071	<u>0.125</u>	<u>0.500</u>	0.500
Utility ↑	1.000	0.951	0.850	0.876	0.880	<u>0.940</u>	0.791	0.698	0.815	0.683
Mean	1.000	0.936	0.848	0.750	0.755	<u>0.873</u>	0.598	0.474	0.609	0.421
Read	1.000	0.922	0.950	0.988	0.991	0.998	0.991	0.983	<u>0.993</u>	0.982
Hall	1.000	0.994	0.752	0.888	0.893	<u>0.947</u>	0.783	0.636	0.844	0.646

Table 2: **Anonymization on the hard dataset**, judged by GLM-5. For SEAL and GRASP we report the best-Overall (_O) and best-privacy (_P) rounds; full trajectories are in Table 9. **Bold/underline**: best/second-best per row.

Metric	Original	Azure	Dipper	Adv. Anon.			SEAL		GRASP (Ours)	
				Gemini	Claude	Qwen3	best-O	best-P	best-O	best-P
Overall ↑	-	-0.023	-0.035	0.301	<u>0.299</u>	0.265	0.266	0.254	0.294	0.281
Privacy ↓	0.781	0.734	0.707	0.465	0.454	0.517	0.420	<u>0.369</u>	0.379	0.332
Age	0.912	0.868	0.789	0.738	0.717	0.800	0.655	<u>0.589</u>	0.608	0.534
Edu	0.834	0.802	0.828	0.665	0.674	0.744	0.616	0.512	<u>0.501</u>	0.417
Gnd	1.000	0.838	0.952	0.738	<u>0.571</u>	0.595	0.667	0.619	0.571	0.500
Inc	0.520	0.541	0.484	0.520	0.529	0.573	0.508	<u>0.467</u>	0.487	0.457
Loc	0.891	0.796	0.761	0.219	0.217	0.227	0.203	<u>0.203</u>	0.212	0.209
Mar	0.907	0.826	0.840	0.662	0.680	0.733	0.667	<u>0.613</u>	0.627	0.527
Occ	0.624	0.566	0.555	0.291	0.265	0.388	0.214	<u>0.147</u>	0.169	0.117
PoB	0.908	0.831	0.806	0.155	0.118	0.168	0.126	0.124	0.126	<u>0.120</u>
Utility ↑	1.000	<u>0.916</u>	0.870	0.896	0.879	0.927	0.804	0.727	0.780	0.706
Mean	1.000	0.898	<u>0.893</u>	0.722	0.678	0.811	0.572	0.449	0.483	0.360
Read	1.000	0.850	0.970	<u>0.995</u>	0.995	0.999	0.993	0.981	0.966	0.947
Hall	1.000	1.000	0.748	<u>0.972</u>	0.966	0.972	0.848	0.750	0.890	0.811

4.3 Ablations and Analysis

Component ablation. Table 3 isolates the contribution of each design choice, starting from the supervised fine-tuned model and toggling one component of the GRPO stage at a time. All variants are evaluated over the first three self-refinement rounds (R0–R3), so that every configuration is compared under an identical protocol; the full model reaches privacy 0.234 on the main split and 0.377 on the hard split. The most important component is the confidence term in the privacy reward: without it, privacy collapses to 0.626/0.773 (main/hard), close to the un-anonymized text (0.620/0.781), since the reward no longer penalizes leaving an attribute exposed as long as the attacker is not fully certain. The truth-aligned reward, which penalizes only correct guesses, leaves main-split privacy

nearly intact (0.252) but degrades the hard split to 0.430: the policy learns to disguise cues just enough to mislead the reference attacker, and the stronger evaluation attacker recovers them. Removing the attacker feedback that conditions each rewrite (0.294/0.387), halving the training data (0.253/0.426), and shrinking the GRPO group to $G=4$ (0.333/0.506) each weaken privacy on both splits, confirming that self-critique, more experience, and richer group comparisons all contribute. Removing the utility floor ($\tau=0$) is the one variant that can improve privacy (0.313 on hard), but it does so by sacrificing meaning: utility falls with every refinement round (0.815→0.650 main, 0.832→0.677 hard), the degenerate trade the floor is designed to prevent. Full per-iteration results for all variants are reported in Table 7.

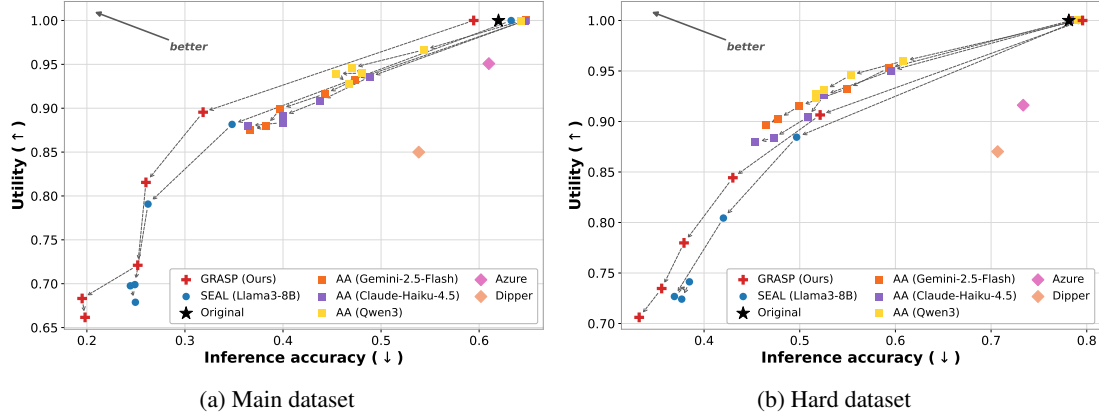


Figure 2: **Privacy-utility trade-off.** Self-refinement trajectories (rounds R0–R5) judged by GLM-5; the upper-left corner is ideal. GRASP extends the Pareto frontier on both splits, reaching the strongest anonymization.

Table 3: **Ablation of GRASP components**, reporting privacy (↓) and utility (↑) at the best-Overall round.

Method	Main		Hard	
	Priv. ↓	Util. ↑	Priv. ↓	Util. ↑
SFT (no RL)	0.388	0.905	0.564	0.917
GRASP (full)	0.234	0.730	0.377	0.773
w/o confidence reward	0.626	0.992	0.773	0.996
w/o attacker feedback	0.294	0.823	0.387	0.761
w/ truth-aligned reward	0.252	0.753	0.430	0.835
w/o utility floor ($\tau=0$)	0.272	0.815	0.313	0.677
fewer samples (2k vs 5k)	0.253	0.749	0.426	0.851
smaller group ($G=4$)	0.333	0.870	0.506	0.888

Table 4: **Judge robustness.** Overall (↑), privacy (↓), and utility (↑) at each method’s best-Overall round under three judges. **Bold**: better per metric.

Judge	Split	SEAL			GRASP (Ours)		
		Ovr.	Priv.	Util.	Ovr.	Priv.	Util.
GLM-5	main	0.368	0.262	0.791	0.396	0.260	0.815
	hard	0.266	0.420	0.804	0.294	0.379	0.780
Qwen3	main	0.342	0.236	0.723	0.400	0.258	0.817
	hard	0.300	0.378	0.784	0.346	0.293	0.721
GPT-OSS	main	0.291	0.317	0.802	0.319	0.309	0.817
	hard	0.299	0.254	0.624	0.357	0.318	0.765

Evaluation with alternative LLM judges. To test whether our conclusions depend on the evaluator, we re-score both splits with two judges from different model families, Qwen3 and GPT-OSS, alongside our primary GLM-5 judge (Table 4). GRASP attains a higher best overall score than SEAL in all six judge×split cells: from 0.342 to 0.400 (main) and 0.300 to 0.346 (hard) under Qwen3, and from 0.291 to 0.319 and 0.299 to 0.357 under GPT-OSS. Absolute values shift between judges, but the ranking is unchanged, and GRASP leads SEAL at nearly every refinement step under every judge (Table 10), trailing only before self-refinement takes effect on the hard split.

Efficiency. Because anonymization runs entirely on a local 8B model, no text leaves the user’s machine. At a 1:1 input–output ratio, the Llama-3.1-8B backbone costs roughly 1% of the GPT-4o teacher, versus 30% for Claude Haiku 4.5 and 14% for Gemini 2.5 Flash. Per-sample rewriting takes 0.64s, versus 0.49–0.81s for the API systems; a full self-refinement step, which adds local attribute inference on the rewrite, takes 13.8s, versus 3.4–5.6s.

Full measurements are in Table 6 in the Appendix.

Qualitative comparison. Appendix E traces how each method rewrites the same low-income comment: frontier anonymizers soften the wording but leave the income cue intact, SEAL over-generalizes into stilted phrasing, and GRASP removes the cue while keeping a natural, coherent sentence.

5 Conclusion

We introduced GRASP, which replaces the offline preference learning of prior anonymizer distillation with online reinforcement learning: a single small model serves as anonymizer, adversary, and utility judge, and GRPO optimizes the privacy–utility objective directly against this self-generated reward, with constraints that guard against reward hacking. On SynthPAI, GRASP improves the privacy–utility trade-off over SEAL under three independent judges, removes more private information than adversarial anonymization driven by frontier models, and extends the Pareto frontier on both evaluation splits—while running entirely on-device at roughly 1% of the teacher’s cost.

Limitations

Inference latency. A full self-refinement step runs attribute inference on the local 8B model, taking 13.8s per sample versus 3.4–5.6s for API-based adversarial anonymization, which offloads this step to a frontier model (Appendix C). This is the price of keeping all computation on-device; plain rewriting is comparable to the API systems (0.64s), and users who need lower latency can run fewer refinement rounds at a reduced privacy level.

LLM-based evaluation. Privacy and utility are scored by LLM judges. Judge behavior is known to be sensitive to the evaluation protocol itself, including position and verbosity effects, and to vary with the task being scored (Hariri et al., 2026; Zheng et al., 2023). We mitigate evaluator-specific artifacts by replicating all comparisons under three judges from different model families (Table 4), but LLM judges remain imperfect proxies, particularly for utility, where the judgment is open-ended rather than checkable against a reference (Ghiasvand et al., 2026).

Fixed attribute set. GRASP is trained and evaluated on the eight personal attributes annotated in SynthPAI; protecting a different or broader attribute set (e.g., health status or political views) would require regenerating trajectories and retraining, and performance on attributes outside this set is untested.

Potential risks. GRASP reduces but does not eliminate attribute leakage, and its privacy measurements reflect current inference models; users should not treat its output as a guarantee of anonymity, particularly for high-stakes disclosures. Conversely, because the model is trained to predict private attributes as part of its self-critique, it could in principle be misused as an inference attacker, although its capability is distilled from, and remains below, that of the publicly available frontier models already capable of such inference.

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A Dataset Details

We build on SynthPAI (Yukhymenko et al., 2024), a corpus of synthetic personal profiles paired with text comments generated from those profiles. We use the 3,456 comments with high-quality human labels for eight personal attributes: age, education level, gender, income level, location, marital status, occupation, and place of birth. Because our contribution is the reinforcement-learning objective rather than the distillation data, we reuse the anonymization trajectories of the SFT+DPO distillation baseline (SEAL) (Kim et al., 2026). These trajectories come from simulating adversarial anonymization (Staab et al., 2025) for up to three steps, with GPT-4o (Achiam et al., 2023) acting as anonymizer, attribute-inference model, and utility evaluator. Of the 300 profiles, 275 are used for trajectory generation (2,734 comments) and 25 are held out entirely from training.

The held-out profiles yield two evaluation splits. The *main* set consists of the 723 comments written from the held-out profiles. The *hard* set consists of 500 texts that embed personal information contextually rather than as explicit identifiers, constructed by SEAL (Kim et al., 2026): hard-to-anonymize texts are first identified in the original SynthPAI pool via iterative adversarial anonymization, and an LLM is then prompted to generate 20 new texts per held-out profile emulating their characteristics—attributes conveyed through narrative and distributed linguistic cues rather than replaceable keywords. Its higher baseline inference accuracy (0.781 vs. 0.620 on the main set) reflects this increased difficulty.

This mirrors the intended use of the framework, in which trajectories are distilled from synthetic profiles using an external LLM, while the resulting small model runs locally on real, private data without invoking an untrusted external service.

B Implementation Details

Unless noted otherwise, GRASP is trained on Llama-3.1-8B-Instruct (Grattafiori et al., 2024). The SFT stage follows the distillation baseline exactly, using LoRA (rank 16, $\alpha_{\text{LoRA}}=16$, dropout 0.05), AdamW at learning rate $2e-4$, and one epoch. We then initialize the policy from the SFT model, which also serves as the frozen reference π_{ref} , and optimize it with GRPO for one epoch on 5,000 training prompts drawn from the trajectory states. Each prompt draws $G=8$ sam-

Table 5: **Hyperparameters for the SFT and GRPO stages of GRASP.** The SFT stage matches SEAL; GRPO is run for a single self-refinement round.

	Parameter	SFT	GRPO
Training	Effective batch size	16	32
	Train epochs / rounds	1	1
	KL weight β_{KL}	-	0.03
	Sampling temperature	-	0.9
	Mixed precision	bf16	bf16
Optimization	Optimizer	AdamW	AdamW
	Learning rate	$2e-4$	$5e-6$
	Weight decay	$1e-2$	$1e-2$
	(β_1, β_2)	(0.9, 0.999)	(0.9, 0.999)
	Max gradient norm	1.0	1.0
LoRA	Max completion length	-	512
	Rank	16	16
	Alpha	16	16
Reward	Dropout	0.05	0.05
	Group size G	-	8
	Privacy coeffs (α, β)	-	(0.7, 0.3)
	Weights (w_p, w_u)	-	(0.9, 0.1)
	Utility floor τ	-	0.55
Reward	Training prompts	-	5,000

pled anonymizations, and training uses learning rate $5e-6$, KL weight $\beta_{\text{KL}}=0.03$, effective batch size 32, sampling temperature 0.9, and a maximum completion length of 512 tokens. The reward uses privacy coefficients $\alpha=0.7$, $\beta=0.3$, trade-off weights $w_p=0.9$, $w_u=0.1$, and utility floor $\tau=0.55$. All models are trained with LoRA in mixed precision with FlashAttention-2 on NVIDIA GPUs. At inference the trained policy anonymizes by self-refinement for up to five iterations. Table 5 summarizes all hyperparameters for both stages.

C Efficiency

Table 6 reports inference cost and per-sample latency, measured over 50 samples with GRASP running on a single NVIDIA GPU and the baselines accessed via their APIs. Cost assumes a 1:1 input-to-output token ratio and is stated relative to the GPT-4o teacher: the Llama-3.1-8B backbone used by GRASP comes to roughly 1%, versus 30% for Claude Haiku 4.5 and 14% for Gemini 2.5 Flash. For rewriting alone, per-sample latency is 0.64s, within the range of the API models (0.49–0.81s). A full self-refinement step additionally runs attribute inference on the rewrite; because GRASP performs this step with the same local 8B model whereas the baselines use their frontier API model, its latency rises to 13.8s versus 3.4–5.6s. Note that the latency comparison is approximate, as the API models run on optimized serving infrastructure while GRASP runs on a single local GPU; the corresponding trade-off is discussed in the Limitations Sec-

Algorithm 1 GRASP

Require: reference model π_{ref} (SFT, §2); group size G ; weights w_p, w_u ; coefficients α, β ; utility floor τ ; KL weight β_{KL}

Ensure: anonymization policy π_θ

```
1: Initialize  $\pi_\theta \leftarrow \pi_{\text{ref}}$ 
2: while not converged do
3:   Sample a text  $x$  with target attribute set  $\mathcal{P}$ 
4:   Draw a group  $\{x'_1, \dots, x'_G\} \sim \pi_{\theta_{\text{old}}}(\cdot \mid x)$ 
5:   for  $i = 1$  to  $G$  do
6:      $\mathcal{P}' \leftarrow \mathcal{M}_{\text{priv}}(x'_i)$   $\triangleright \pi_{\text{ref}}$  as adversary
7:      $r_p \leftarrow \alpha(1 - \frac{|\mathcal{P}'|}{|\mathcal{P}|}) + \beta(1 - \overline{\text{conf}})$   $\triangleright \text{Eq. (3)}$ 
8:      $r_u \leftarrow \mathcal{M}_{\text{util}}(x'_i, x)$   $\triangleright \pi_{\text{ref}}$  as evaluator
9:     if  $r_u \geq \tau$  and  $|x'_i| \in [\frac{1}{2}|x|, 2|x|]$  then
10:       $R_i \leftarrow w_p r_p + w_u r_u$ 
11:     else
12:       $R_i \leftarrow -1$   $\triangleright$  infeasible rewrite
13:     end if
14:   end for
15:    $\hat{A}_i \leftarrow (R_i - \text{mean}(\{R_j\})) / \text{std}(\{R_j\})$  for all  $i$ 
16:   Update  $\theta$  by maximizing  $\mathcal{J}(\theta)$  with KL to  $\pi_{\text{ref}}$   $\triangleright \text{Eq. (5)}$ 
17: end while
18: return  $\pi_\theta$ 
```

Table 6: **Efficiency.** Inference cost (per 1M tokens; relative to the GPT-4o teacher at a 1:1 token ratio) and per-sample latency, measured over 50 samples. GRASP runs locally.

Model	Cost / 1M		Rel.	Latency (s)	
	In	Out		Anon.	Anon.+Inf.
Claude Haiku 4.5	\$1.00	\$5.00	30.0%	0.81	5.64
Gemini 2.5 Flash	\$0.30	\$2.50	14.0%	0.49	3.37
GRASP / SEAL	\$0.10	\$0.10	1.0%	0.64	13.76

tion.

D Per-iteration results

Main and Hard splits. Tables 8 and 9 report the full self-refinement trajectories summarized in Tables 1 and 2. Two patterns hold across methods and splits. First, privacy improves with refinement while utility declines monotonically, so the overall score peaks at an intermediate round—for GRASP, round 2 on both splits (0.396 main, 0.294 hard)—and the per-round trajectory lets users pick an operating point along this trade-off. Second, the methods differ in how far refinement carries them: the frontier adversarial anonymizers improve privacy slowly and remain well above 0.36 (main) and 0.45 (hard) after five rounds, whereas GRASP

continues to gain, reaching 0.195 on the main set (round 4) and 0.332 on the hard set (round 5). On the hard split GRASP’s privacy decreases at every round, and its per-attribute gains over SEAL concentrate on the same attributes as on the main split, consistent with the aggregate results in Section 4.2.

Ablation. Table 7 shows how each ablated variant evolves over refinement, revealing dynamics that the best-round summary in Table 3 compresses. Without the confidence reward, refinement makes no progress at all: privacy stays at 0.626–0.643 (main) and 0.773–0.785 (hard) across all three rounds, and the overall score never rises above 0.006. The utility floor shapes the trajectory rather than a single round: with $\tau=0$ the policy reaches stronger raw privacy at every matched round, but utility falls steeply with each refinement (0.815→0.650 main, 0.832→0.677 hard) and the overall score declines from R1 to R3 (0.376→0.303 on main), whereas the full model’s overall score rises (0.282→0.353)—the floor trades some immediate privacy for refinement that remains worthwhile. The remaining variants—no attacker feedback, fewer training samples, and the smaller group $G=4$ —track the full model’s shape but sit at consistently weaker privacy, with $G=4$ the furthest behind at every round.

Table 7: **Per-iteration ablation of GRASP components.** Overall ($\uparrow$), privacy ($\downarrow$, attribute-inference accuracy), and utility ($\uparrow$) at self-refinement rounds R1–R3 (GLM-5 judge); R0 is the un-anonymized input and is omitted. Highlighted: full model. Summarized at the best-Overall round in Table 3.

Method	Ovr. $\uparrow$			Priv. $\downarrow$			Util. $\uparrow$		
	R1	R2	R3	R1	R2	R3	R1	R2	R3
<i>Main dataset</i>									
SFT (no RL)	0.172	0.216	0.279	0.483	0.436	0.388	0.951	0.919	0.905
GRASP (full)	0.282	0.351	0.353	0.378	0.281	0.234	0.892	0.804	0.730
w/o confidence reward	-0.038	-0.017	-0.045	0.639	0.626	0.643	0.993	0.992	0.991
w/o attacker feedback	0.348	0.336	0.308	0.294	0.246	0.239	0.823	0.734	0.693
w/ truth-aligned reward	0.276	0.324	0.346	0.383	0.301	0.252	0.893	0.809	0.753
w/o utility floor ($\tau=0$)	0.376	0.360	0.303	0.272	0.212	0.215	0.815	0.702	0.650
fewer samples (2k vs 5k)	0.296	0.319	0.341	0.373	0.306	0.253	0.898	0.813	0.749
smaller group ($G=4$)	0.212	0.299	0.333	0.453	0.373	0.333	0.943	0.900	0.870
<i>Hard dataset</i>									
SFT (no RL)	0.133	0.169	0.195	0.641	0.596	0.564	0.953	0.932	0.917
GRASP (full)	0.242	0.278	0.290	0.519	0.436	0.377	0.906	0.837	0.773
w/o confidence reward	-0.006	0.003	0.006	0.785	0.776	0.773	0.999	0.997	0.996
w/o attacker feedback	0.249	0.265	0.262	0.478	0.387	0.356	0.861	0.761	0.717
w/ truth-aligned reward	0.229	0.285	0.273	0.521	0.430	0.395	0.896	0.835	0.779
w/o utility floor ($\tau=0$)	0.266	0.256	0.276	0.443	0.370	0.313	0.832	0.730	0.677
fewer samples (2k vs 5k)	0.253	0.306	0.300	0.510	0.426	0.382	0.907	0.851	0.789
smaller group ($G=4$)	0.191	0.238	0.239	0.586	0.524	0.506	0.940	0.908	0.888

Judge robustness. Table 10 extends the best-round comparison of Table 4 to every self-refinement step. GRASP leads SEAL on the overall score in all but three of the 36 judge $\times$ split $\times$ round cells: the first round on the hard split under GLM-5 (0.239 vs. 0.248) and Qwen3 (0.244 vs. 0.253), before self-refinement takes effect, and the final round on the main split under GPT-OSS, where the two methods are within 0.003. Absolute scores shift across judges, mainly through stricter utility assessments, but the per-round ordering is preserved, indicating that the advantage is neither an artifact of the evaluator nor of selecting a favorable round.

E Qualitative comparison

Table 11 traces how each method rewrites a single comment, “rent eats up most of my paycheck these days”, whose author has low income. Azure finds no explicit identifier and leaves the text unchanged. Dipper paraphrases without an anonymization objective and distorts the meaning (“My rent goes a long way these days”), consistent with its large utility loss in Tables 8 and 9. The frontier adversarial anonymizers soften “paycheck” into “income” or “expenses” but still convey that the cost of living strains the author, leaving the low-income cue largely intact. SEAL over-generalizes across rounds, drifting into stilted, near-meaningless phrasing (“a significant amount

of funds is being allocated to a particular area”). GRASP instead removes the personal-income cue while keeping a natural, coherent sentence (“It can be expensive to pay for a place to live”). This mirrors the quantitative picture: GRASP reaches the strongest privacy while preserving readable, faithful text.

Table 8: **Per-iteration anonymization results on the main dataset.** Overall ($\uparrow$), privacy ($\downarrow$, attribute-inference accuracy), and utility ($\uparrow$) per self-refinement round, judged by GLM-5, with per-attribute inference accuracy. Overall is the relative privacy improvement minus the relative utility loss, normalized by the original privacy ($P_{\text{orig}}=0.620$). Top panel: the five-round trajectories of the three adversarial-anonymization baselines; bottom panel: the original text, Azure, Dipper, and the trajectories of SEAL and GRASP. **Bold/underline**: best/second-best per row. GRASP attains both the best overall score (0.396, round 2) and the lowest inference accuracy (0.195, round 4). Summarized in Table 1 and plotted in Figure 2.

Metric	Adv. Anon. Gemini					Adv. Anon. Claude					Adv. Anon. Qwen3				
	iter 1	iter 2	iter 3	iter 4	iter 5	iter 1	iter 2	iter 3	iter 4	iter 5	iter 1	iter 2	iter 3	iter 4	iter 5
Overall $\uparrow$	0.169	0.202	0.259	0.262	0.284	0.148	0.202	0.246	0.239	0.292	0.089	0.188	0.165	0.207	0.175
Privacy $\downarrow$	0.473	0.443	0.397	0.383	0.367	0.488	0.438	0.400	0.400	0.365	0.544	0.470	0.480	0.454	0.467
Age	0.700	0.703	0.693	0.703	0.720	0.743	0.713	0.683	0.713	0.614	0.733	0.733	0.733	0.750	0.752
Edu	0.640	0.635	0.576	0.550	0.521	0.630	0.621	0.514	0.531	0.510	0.654	0.602	0.616	0.576	0.619
Gnd	0.639	0.623	0.525	0.525	0.508	0.672	0.541	0.607	0.656	0.590	0.639	0.607	0.672	0.623	0.689
Inc	0.541	0.592	0.571	0.561	0.510	0.571	0.561	0.510	0.541	0.490	0.653	0.561	0.571	0.561	0.592
Loc	0.127	0.082	0.060	0.067	0.067	0.179	0.142	0.127	0.114	0.090	0.209	0.134	0.127	0.097	0.112
Mar	0.662	0.627	0.560	0.587	0.554	0.653	0.627	0.613	0.640	0.608	0.747	0.667	0.653	0.693	0.680
Occ	0.320	0.246	0.197	0.173	0.163	0.333	0.249	0.223	0.185	0.167	0.429	0.325	0.334	0.289	0.280
PoB	0.179	0.214	0.214	0.107	0.143	0.143	0.179	0.179	0.214	0.179	0.143	0.071	0.071	0.143	0.143
Utility $\uparrow$	0.932	0.916	0.899	0.880	0.876	0.936	0.908	0.891	0.884	0.880	0.967	0.946	0.940	0.940	0.928
Mean	0.893	0.849	0.804	0.774	0.750	0.889	0.824	0.793	0.768	0.755	0.930	0.898	0.879	0.873	0.861
Read	0.998	0.995	0.993	0.989	0.988	0.998	0.995	0.993	0.994	0.991	0.999	0.998	0.997	0.998	0.998
Hall	0.906	0.905	0.900	0.876	0.888	0.920	0.906	0.888	0.888	0.893	<u>0.971</u>	0.942	0.943	0.947	0.924

Metric	Original	Azure	Dipper	SEAL					GRASP (Ours)				
				iter 1	iter 2	iter 3	iter 4	iter 5	iter 1	iter 2	iter 3	iter 4	iter 5
Overall $\uparrow$	-	-0.033	-0.018	0.320	0.368	0.297	0.304	0.276	<u>0.382</u>	0.396	0.315	0.368	0.342
Privacy $\downarrow$	0.620	0.610	0.538	0.348	0.262	0.249	0.244	0.249	0.319	0.260	0.252	0.195	<u>0.198</u>
Age	0.762	0.802	0.614	0.624	0.551	0.567	<u>0.500</u>	0.538	0.550	0.600	0.650	0.300	0.500
Edu	0.645	0.667	0.718	0.502	0.416	0.313	0.325	0.345	0.391	0.266	0.297	0.203	<u>0.203</u>
Gnd	0.883	0.820	0.721	0.607	0.450	0.345	0.417	0.386	0.583	0.417	0.333	0.250	<u>0.250</u>
Inc	0.561	0.598	0.531	0.485	0.457	0.559	0.521	0.516	0.571	0.607	0.536	0.536	<u>0.464</u>
Loc	0.470	0.362	0.300	0.030	0.015	<u>0.023</u>	0.023	0.033	0.100	0.050	0.050	0.050	0.050
Mar	0.635	0.676	0.541	0.600	0.324	0.479	0.425	0.384	0.273	0.409	0.318	0.182	<u>0.182</u>
Occ	0.578	0.542	0.448	0.139	0.069	0.045	<u>0.048</u>	0.055	0.168	0.069	0.082	0.083	0.073
PoB	0.321	0.286	0.167	0.071	<u>0.071</u>	0.107	0.125	0.107	0.375	0.500	0.125	0.500	0.500
Utility $\uparrow$	1.000	<u>0.951</u>	0.850	0.882	0.791	0.699	0.698	0.679	0.895	0.815	0.721	0.683	0.662
Mean	1.000	0.936	0.848	0.771	0.598	0.470	0.474	0.453	0.765	0.609	0.496	0.421	0.370
Read	1.000	0.922	0.950	0.997	0.991	0.984	0.983	0.954	0.995	0.993	0.985	0.982	0.974
Hall	1.000	0.994	0.752	0.876	0.783	0.643	0.636	0.630	0.927	0.844	0.682	0.646	0.641

Table 9: **Per-iteration anonymization results on the hard dataset.** Overall ($\uparrow$), privacy ($\downarrow$, attribute-inference accuracy), and utility ($\uparrow$) per self-refinement round, judged by GLM-5, with per-attribute inference accuracy. Overall is the relative privacy improvement minus the relative utility loss, normalized by the original privacy ($P_{\text{orig}}=0.781$). Top panel: the five-round trajectories of the three adversarial-anonymization baselines; bottom panel: the original text, Azure, Dipper, and the trajectories of SEAL and GRASP. **Bold/underline**: best/second-best per row. GRASP’s inference accuracy decreases at every round, reaching 0.332, the lowest of any method. Summarized in Table 2 and plotted in Figure 2.

Metric	Adv. Anon. Gemini					Adv. Anon. Claude					Adv. Anon. Qwen3				
	iter 1	iter 2	iter 3	iter 4	iter 5	iter 1	iter 2	iter 3	iter 4	iter 5	iter 1	iter 2	iter 3	iter 4	iter 5
Overall $\uparrow$	0.194	0.229	0.276	0.291	0.301	0.188	0.253	0.253	0.278	<u>0.299</u>	0.181	0.238	0.258	0.265	0.262
Privacy $\downarrow$	0.593	0.550	0.499	0.478	0.465	0.595	0.526	0.508	0.473	0.454	0.608	0.553	0.526	0.517	0.517
Age	0.886	0.847	0.780	0.750	0.738	0.868	0.813	0.796	0.751	0.717	0.899	0.836	0.818	0.800	0.811
Edu	0.825	0.785	0.736	0.704	0.665	0.827	0.757	0.719	0.676	0.674	0.821	0.788	0.765	0.744	0.755
Gnd	0.690	0.762	0.714	0.643	0.738	0.738	0.643	0.585	0.643	0.571	0.738	0.619	0.643	0.595	0.610
Inc	0.540	0.607	0.546	0.503	0.520	0.543	0.543	0.564	0.545	0.529	0.547	0.562	0.538	0.573	0.566
Loc	0.280	0.230	0.221	0.219	0.219	0.297	0.245	0.228	0.219	0.217	0.317	0.247	0.229	0.227	0.235
Mar	0.800	0.770	0.747	0.747	0.662	0.800	0.733	0.787	0.716	0.680	0.813	0.773	0.760	0.733	0.743
Occ	0.541	0.413	0.323	0.318	0.291	0.534	0.387	0.368	0.300	0.265	0.575	0.489	0.421	0.388	0.357
PoB	0.299	0.190	0.179	0.162	0.155	0.321	0.218	0.166	0.140	0.118	0.300	0.187	0.160	0.168	0.170
Utility $\uparrow$	<u>0.953</u>	0.932	0.915	0.903	0.896	0.950	0.926	0.904	0.884	0.879	0.960	0.946	0.931	0.927	0.924
Mean	0.879	0.821	0.775	0.754	0.722	0.873	0.799	0.753	0.699	0.678	0.906	0.861	0.830	0.811	0.794
Read	0.999	0.998	0.999	0.998	0.995	1.000	0.999	0.998	0.998	0.995	<u>1.000</u>	0.999	0.999	0.999	0.999
Hall	<u>0.980</u>	0.978	0.972	0.956	0.972	0.978	0.980	0.962	0.954	0.966	0.974	0.978	0.964	0.972	0.978

Metric	Original	Azure	Dipper	SEAL					GRASP (Ours)				
				iter 1	iter 2	iter 3	iter 4	iter 5	iter 1	iter 2	iter 3	iter 4	iter 5
Overall $\uparrow$	-	-0.023	-0.035	0.248	0.266	0.254	0.249	0.242	0.239	0.294	0.294	0.279	0.281
Privacy $\downarrow$	0.781	0.734	0.707	0.497	0.420	0.369	0.385	0.377	0.521	0.430	0.379	<u>0.356</u>	0.332
Age	0.912	0.868	0.789	0.744	0.655	0.589	0.623	0.606	0.804	0.679	0.608	<u>0.580</u>	0.534
Edu	0.834	0.802	0.828	0.714	0.616	0.512	0.527	0.550	0.723	0.593	0.501	<u>0.472</u>	0.417
Gnd	1.000	0.838	0.952	0.738	0.667	0.619	0.571	0.595	0.810	0.595	0.571	0.500	<u>0.500</u>
Inc	0.520	0.541	0.484	0.538	0.508	<u>0.467</u>	0.506	0.470	0.552	0.540	0.487	0.475	0.457
Loc	0.891	0.796	0.761	0.240	0.203	0.203	0.200	<u>0.201</u>	0.273	0.221	0.212	0.208	0.209
Mar	0.907	0.826	0.840	0.693	0.667	0.613	0.627	0.589	0.760	0.613	0.627	<u>0.560</u>	0.527
Occ	0.624	0.566	0.555	0.360	0.214	0.147	0.155	0.137	0.339	0.217	0.169	<u>0.124</u>	0.117
PoB	0.908	0.831	0.806	0.184	0.126	0.124	0.123	0.125	0.241	0.165	0.126	0.126	<u>0.120</u>
Utility $\uparrow$	1.000	0.916	0.870	0.884	0.804	0.727	0.741	0.724	0.906	0.844	0.780	0.735	0.706
Mean	1.000	<u>0.898</u>	0.893	0.727	0.572	0.449	0.457	0.435	0.755	0.600	0.483	0.414	0.360
Read	1.000	<u>0.850</u>	0.970	0.998	0.993	0.981	0.980	0.969	0.996	0.985	0.966	0.954	0.947
Hall	1.000	1.000	0.748	0.928	0.848	0.750	0.786	0.768	0.968	0.948	0.890	0.836	0.811

Table 10: **Per-iteration judge robustness.** Overall ($\uparrow$), privacy ($\downarrow$, attribute-inference accuracy), and utility ($\uparrow$) at each of six self-refinement rounds for SEAL and GRASP, scored by three independent judges on both splits. Bold: better of the two methods per iteration and metric. GRASP leads on Overall at nearly every round under every judge. Summarized at the best-Overall round in Table 4.

Metric	Judge	Method	it1	it2	it3	it4	it5	it6
Main dataset								
Overall ↑	GLM-5	SEAL	0.320	0.368	0.297	0.304	0.276	0.227
		GRASP	0.382	0.396	0.315	0.368	0.342	0.293
	Qwen3	SEAL	0.249	0.320	0.342	0.321	0.300	0.268
		GRASP	0.319	0.400	0.390	0.366	0.384	0.376
	GPT-OSS	SEAL	0.291	0.257	0.222	0.221	0.200	0.202
		GRASP	0.319	0.312	0.237	0.245	0.208	0.199
Privacy ↓	GLM-5	SEAL	0.348	0.262	0.249	0.244	0.249	0.268
		GRASP	0.319	0.260	0.252	0.195	0.198	0.207
	Qwen3	SEAL	0.375	0.286	0.236	0.246	0.245	0.255
		GRASP	0.347	0.258	0.234	0.227	0.202	0.194
	GPT-OSS	SEAL	0.317	0.254	0.214	0.217	0.222	0.211
		GRASP	0.309	0.243	0.232	0.198	0.195	0.181
Utility ↑	GLM-5	SEAL	0.882	0.791	0.699	0.698	0.679	0.660
		GRASP	0.895	0.815	0.721	0.683	0.662	0.627
	Qwen3	SEAL	0.853	0.782	0.723	0.718	0.694	0.679
		GRASP	0.879	0.817	0.767	0.733	0.710	0.689
	GPT-OSS	SEAL	0.802	0.667	0.567	0.570	0.558	0.542
		GRASP	0.817	0.704	0.611	0.564	0.523	0.491
Hard dataset								
Overall ↑	GLM-5	SEAL	0.248	0.266	0.254	0.249	0.242	0.232
		GRASP	0.239	0.294	0.294	0.279	0.281	0.264
	Qwen3	SEAL	0.253	0.300	0.285	0.297	0.273	0.264
		GRASP	0.244	0.327	0.336	0.335	0.346	0.329
	GPT-OSS	SEAL	0.286	0.296	0.299	0.281	0.272	0.266
		GRASP	0.306	0.357	0.352	0.308	0.298	0.274
Privacy ↓	GLM-5	SEAL	0.497	0.420	0.369	0.385	0.377	0.376
		GRASP	0.521	0.430	0.379	0.356	0.332	0.326
	Qwen3	SEAL	0.472	0.378	0.340	0.329	0.340	0.338
		GRASP	0.497	0.388	0.340	0.318	0.293	0.283
	GPT-OSS	SEAL	0.417	0.326	0.254	0.268	0.269	0.266
		GRASP	0.425	0.318	0.270	0.254	0.230	0.216
Utility ↑	GLM-5	SEAL	0.884	0.804	0.727	0.741	0.724	0.714
		GRASP	0.906	0.844	0.780	0.735	0.706	0.682
	Qwen3	SEAL	0.858	0.784	0.720	0.718	0.708	0.696
		GRASP	0.881	0.824	0.772	0.742	0.721	0.691
	GPT-OSS	SEAL	0.820	0.713	0.624	0.624	0.617	0.607
		GRASP	0.850	0.765	0.697	0.633	0.593	0.550

Table 11: **Round-by-round anonymization of the same comment by each method**, where the goal is to hide that the author has low income (their rent consumes most of their paycheck). Azure leaves the text unchanged and Dipper paraphrases but distorts the meaning; the frontier adversarial anonymizers soften the wording but keep the financial-strain signal; SEAL over-generalizes into stilted, near-meaningless phrasing, whereas GRASP removes the income cue while keeping a natural, coherent sentence. Identical consecutive rounds are merged.

Method	Round	Text
Original	–	rent eats up most of my paycheck these days
Azure	1–5	rent eats up most of my paycheck these days
Dipper	1	My rent goes a long way these days
AA (Gemini)	1	Rent eats up most of my income these days
	2	Housing costs eat up most of my income these days
	3–4	Housing costs are a significant expense these days
	5	The cost of living is a significant expense these days
AA (Claude)	1	rent eats up a significant portion of my paycheck these days
	2	housing costs take up a significant portion of my paycheck these days
	3–5	housing costs take up a significant portion of my expenses these days
AA (Qwen3)	1–5	Housing costs eat up most of my income these days
SEAL	1	rent takes up a lot of money right now
	2	a lot of money is being spent on something right now
	3–4	a lot of money is being spent on something
	5	a significant amount of funds is being allocated to a particular area
GRASP	1	a lot of money is going towards housing
	2	a lot of money is going towards housing costs
	3	Housing costs are a significant expense.
	4	It can be expensive to pay for a place to live.
	5	It can be expensive to find a place to live.